QUANTUMVAULT RESEARCH ARCHIVE & PEER-REVIEWED REFERENCE MONOGRAPH

Category: ERRORCORRECTION | Speedup: Foundational | Concept ID #40

Foundational Research & Advances in Quantum Volume (QV)

Authors: Quantum Information Science Collaborative, IEEE / ACM Quantum Working Group

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Quantum Volume (QV) in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"A horsepower test for quantum computers: it measures not just how many qubits you have, but how good your gates, connections, and error rates actually are."

3. Mathematical Formulation & Unitary Rule

$$V_Q = 2^{m^*}, \quad h_{\text{measured}} > \frac{2}{3}, \quad h = \sum_{x \in U_H} P(x)$$

m^* is the largest circuit width and depth where heavy output probability h exceeds 2/3.

5. Executable IBM Qiskit (Python) Code

```python
from qiskit.circuit.library import QuantumVolume

# 4-qubit Quantum Volume circuit (depth = 4)
qv_circuit = QuantumVolume(num_qubits=4, depth=4)
print("QV circuit ready for heavy output benchmarking.")
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
O(2^m) to compute heavy output bitstrings classically for verification	Hardware benchmark metric scaling exponentially with achievable circuit	Foundational

7. Key Applications & Future Research Directions

- Commercial quantum processor comparison (IBM, Quantinuum, IonQ)
- Evaluating cross-talk, compiler optimization, and gate fidelities
- Tracking hardware roadmap milestones toward quantum advantage

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant ErrorCorrection pipelines.